Interactive Oracle Proofs of Proximity

October 29, 2026

Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book



Plan

1 IOPs of Proximity

2 Compiling Poly-IOPs using RS-IOPPs

3 Batching RS-IOPPs

4 Attempt #3: Putting Everything Together

What will we learn today?

- Recall: a **Poly-IOP** becomes a SNARK by (1) replacing polynomials with **polynomial commitments** and (2) applying **Fiat–Shamir**.
- PlonK = Poly-IOP + KZG. Replacing KZG with Bulletproofs gives Halo2.
- Today: commitments that need only **hash functions**, based on Reed–Solomon codes.

💬 *Key idea*

An RS-IOPP (Reed–Solomon IOP of Proximity) can be used as a polynomial commitment scheme, which turns a Poly-IOP into an IOP.

💬 *Recall from the previous lecture*

$\mathrm{RS}[\mathbb{F}, \Omega, k] := \{\bar{f} : \Omega \to \mathbb{F},\ f \in \mathbb{F}[T]^{<k}\}$, rate $\rho = k/n$, and the list $\mathrm{List}[\mathbf{w}, C, \delta]$ of codewords δ-close to $\mathbf{w}$.

IOPs of Proximity

Interactive Oracle Proofs

Recall the idea of an **interactive oracle proof** (IOP):

- As in any interactive proof, $\mathcal{P}$ and $\mathcal{V}$ exchange messages over several rounds; the verifier's messages are random challenges.
- The prover's messages are **oracles**: long strings that the verifier does not read in full, but only **queries** at a few positions.
- Hence the verifier runs in time much smaller than the total length of the prover's messages.
- In practice, the prover commits to each oracle with a **Merkle tree**, and every query is answered with an opening.

Interactive Oracle Proofs

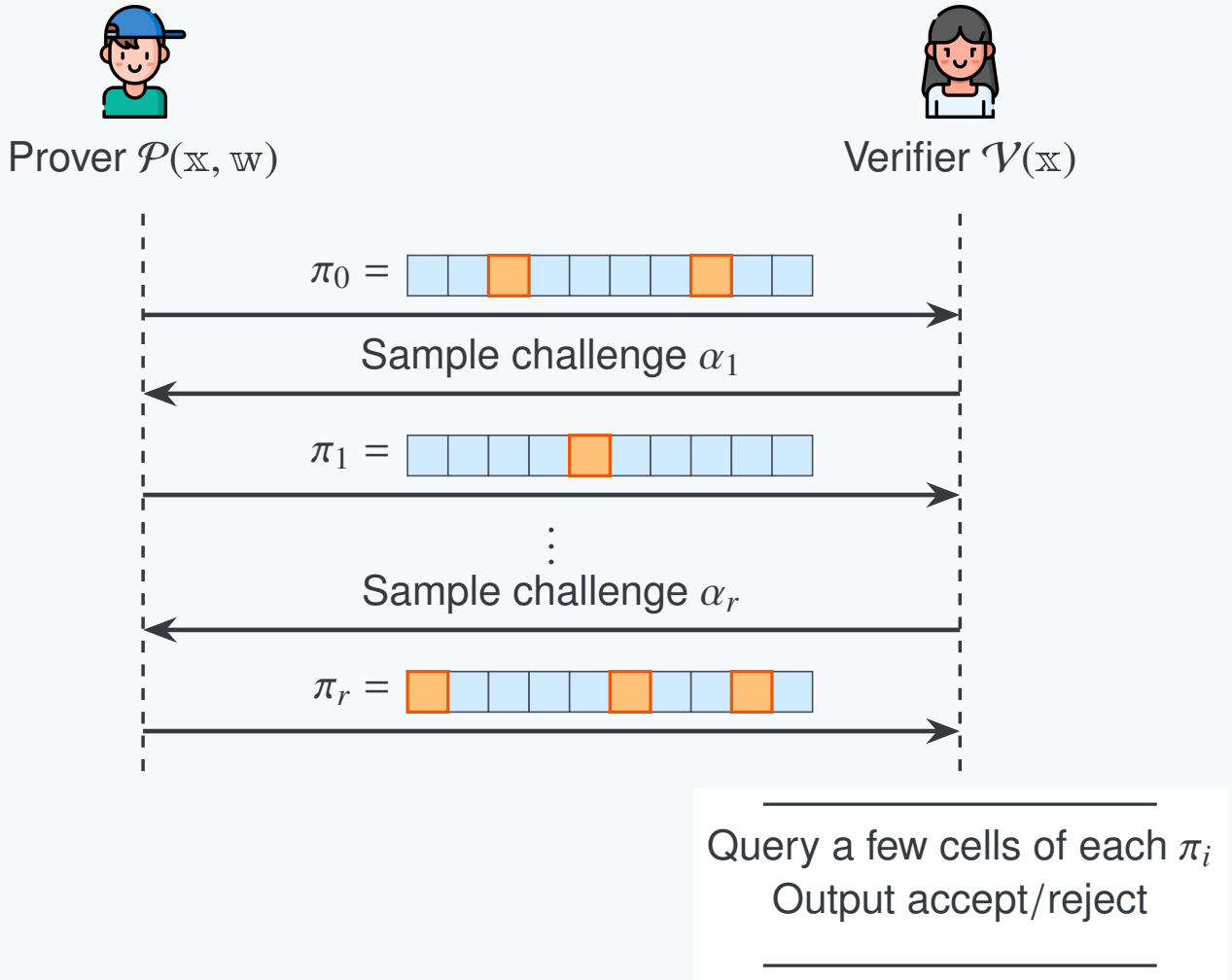


Figure: Long oracles π_i, short challenges α_i; the verifier queries a few cells (orange).

IOP of Proximity

In an IOP, the verifier reads the statement $\mathbb{x}$ in full. For the proximity problem this is not enough: the word $\mathbf{w}$ we want to check is itself too long to read.

Definition (IOP of Proximity)

Let $\mathcal{R} = \{(\mathbb{x}, \mathbb{y}, \mathbb{w})\}$ be a relation, where $\mathbb{y}$ is a string of large length n. An **IOP of proximity** (IOPP) for $\mathcal{R}$ is an IOP in which

- the prover receives $(\mathbb{x}, \mathbb{y}, \mathbb{w})$;
- the verifier receives $\mathbb{x}$ and only **oracle access** to $\mathbb{y}$.

At the end, the verifier outputs

$$\mathcal{V}^{\mathbb{y},\pi_0,\ldots,\pi_r}(\mathbb{x}, \alpha_1, \ldots, \alpha_r) \to \{0, 1\}.$$

IOP of Proximity

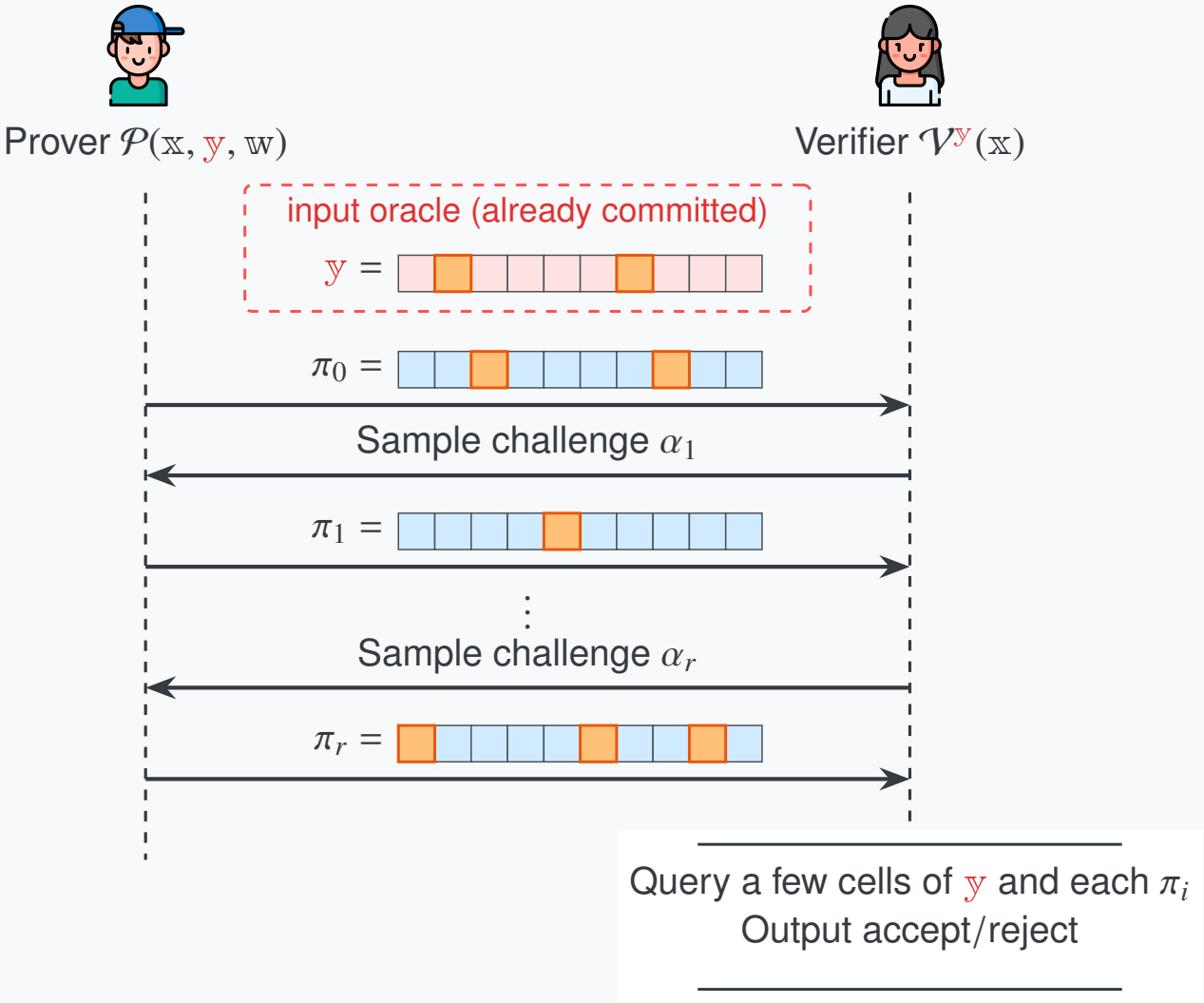


Figure: The input oracle $\mathbb{y}$: known to $\mathcal{P}$ in full, only queried by $\mathcal{V}$.

Which Inputs Should be Rejected?

The natural answer, "every $\mathbb{y}$ that is not valid", is unachievable:

- take a valid $\mathbb{y}$ and change a single cell;
- a verifier querying q of the n cells reads the changed one with probability at most q/n.

We only ask to reject oracles that are far from every valid one.

Definition

$(\mathbb{x}, \mathbb{y})$ is δ**-far from** $\mathcal{R}$ if

$$(\mathbb{x}, \mathbb{y}', \mathbb{w}) \notin \mathcal{R} \text{ for all } \mathbb{y}', \mathbb{w} \text{ such that } \delta(\mathbb{y}, \mathbb{y}') \leq \delta.$$

That is, $\mathbb{y}$ cannot be made valid by changing at most a δ fraction of its cells.

Correctness and Soundness

Definition

An IOPP $(\mathcal{P}, \mathcal{V})$ for $\mathcal{R}$ may have the following properties:

- **Correctness.** For all $(\mathbb{x}, \mathbb{y}, \mathbb{w}) \in \mathcal{R}$, $\mathcal{V}$ always accepts $\mathcal{P}$.
- δ**-soundness.** For all $(\mathbb{x}, \mathbb{y})$ that are δ-far from $\mathcal{R}$,

$$\forall_{\mathcal{P}^*} \ \Pr[\mathcal{V}^{\mathbb{y},\pi_0,\ldots,\pi_r}(\mathbb{x}, \alpha_1, \ldots, \alpha_r) = 1] < \mathsf{negl}(\lambda).$$

The gap is harmless

An invalid $\mathbb{y}$ that is not δ-far may be accepted. But then $\mathbb{y}$ is δ-close to some valid $\mathbb{y}'$, which for a code and small δ is unique, or one of a bounded list (list decoding!). We treat the prover as committed to $\mathbb{y}'$.

Example

Example

Let $\mathbb{x} = \mathsf{RS}[\mathbb{F}_7, \Omega := \mathbb{F}_7, k := 2]$, and $(\mathbb{x}, \mathbb{y}, \mathbb{w}) \in \mathcal{R}$ iff $\mathbb{w} = f \in \mathbb{F}_7[T]^{<2}$ and $\mathbb{y} = \bar{f}$. The valid oracles are exactly the codewords.

- $\mathbb{y}_1 = (3, 5, 0, 2, 4, 6, 1)$ is the codeword of $f = 3 + 2T$: always accepted.
- $\mathbb{y}_2 = (3, 5, 0, 2, 4, 6, 0)$ is $1/7$-close to $\mathbb{y}_1$, so it is not δ-far for $\delta \geq 1/7$: may be accepted, and we regard the prover as committed to $f = 3 + 2T$.
- $\mathbb{y}_3 = (0, 1, 4, 2, 2, 4, 1)$ are the evaluations of T^2: it differs from every codeword in at least 5 cells, so it is δ-far for every $\delta < 5/7$, and it is rejected with high probability.

Reed–Solomon IOPP

The special case where the statement is a code $C = \mathsf{RS}[\mathbb{F}, \Omega, k]$ and $\mathbb{y}$ is a restriction $g : \Omega \to \mathbb{F}$ is the central topic of this lecture.

Definition (Reed–Solomon IOPP)

Given oracle access to $g : \Omega \to \mathbb{F}$ and $\delta \in (0, 1)$, distinguish, with high confidence and few oracle queries $\pi_0, \ldots, \pi_r$, between

- the case $g \in \mathsf{RS}[\mathbb{F}, \Omega, k]$, and
- the case that g is δ-far from $\mathsf{RS}[\mathbb{F}, \Omega, k]$.

Black box

Today we use a δ-RS-IOPP as a black box. Next lecture: how to build one (FRI).

Compiling Poly-IOPs using RS-IOPPs

Attempt #1: Encoding Polynomials

An IOP sends large **strings**, not **polynomials**. How do we encode $f \in \mathbb{F}[T]^{<k}$ into a string of length n?

- Natural answer: the restriction $\bar{f} : \Omega \to \mathbb{F}$, which lies in $C := \mathsf{RS}[\mathbb{F}, \Omega, k]$.
- But the prover can send an arbitrary $g : \Omega \to \mathbb{F}$, so we need to verify that g is close to C: exactly an RS-IOPP!

That is not enough

A polynomial commitment needs **evaluation proofs**: that $f(r) = y$ for a given (r, y).

Quotienting

Proposition (Quotienting)

Let $r \in \mathbb{F} \setminus \Omega$, $y \in \mathbb{F}$, $f \in \mathbb{F}[T]^{<k}$, $\delta \in [0,1]$. Define the **quotient map** q sending a restriction $g : \Omega \to \mathbb{F}$ to the restriction $(g(T) - y)/(T - r)$. Then:

1. if $g = \overline{f} \in \mathrm{RS}[\mathbb{F}, \Omega, k]$ and $y = f(r)$, then $\mathrm{q}(g) \in \mathrm{RS}[\mathbb{F}, \Omega, k-1]$;
2. if $y \neq h(r)$ for all $\overline{h} \in \mathrm{List}[g, k, \delta]$, then $\mathrm{q}(g)$ is δ-far from $\mathrm{RS}[\mathbb{F}, \Omega, k-1]$.

Here, $\mathrm{List}[g, k, \delta] := \mathrm{List}[g, \mathrm{RS}[\mathbb{F}, \Omega, k], \delta]$. Since $r \notin \Omega$, $\mathrm{q}(g)$ is well-defined.

Proof idea.

(i) $f - y$ has a root at r, so $u := (f - y)/(T - r) \in \mathbb{F}[T]^{<k-1}$ and $\mathrm{q}(g) = \overline{u}$.

(ii) If $\mathrm{q}(g)$ is δ-close to some $\overline{u}$, then $h := u \cdot (T - r) + y$ satisfies $h(r) = y$ and $\overline{h} \in \mathrm{List}[g, k, \delta]$.

□

Quotienting: Example

Example

Let $\mathbb{F} = \mathbb{F}_7$, $\Omega = \mathbb{F}_7 \setminus \{0\}$, $k = 2$ and $r = 0 \notin \Omega$. Take $f = 3 + 2T$, so that $g = \bar{f} = (5, 0, 2, 4, 6, 1)$ and $f(r) = 3$.

- ✓ Correct value $y = 3$:

$$\mathrm{q}(g)(\omega) = \frac{3 + 2\omega - 3}{\omega} = 2, \quad \mathrm{q}(g) = (2, 2, 2, 2, 2, 2) \in \mathrm{RS}[\mathbb{F}_7, \Omega, 1].$$

- ✗ Wrong value $y = 4$: $\mathrm{q}(g)(\omega) = 2 - \omega^{-1}$, that is, $\mathrm{q}(g) = (1, 5, 4, 0, 6, 3)$. All six values are distinct, so it is at relative distance $5/6$ from the constants.

This matches the second claim: for small δ, $\mathrm{List}[g, 2, \delta] = \{\bar{f}\}$, and $f(0) = 3 \neq 4$.

Attempt #1

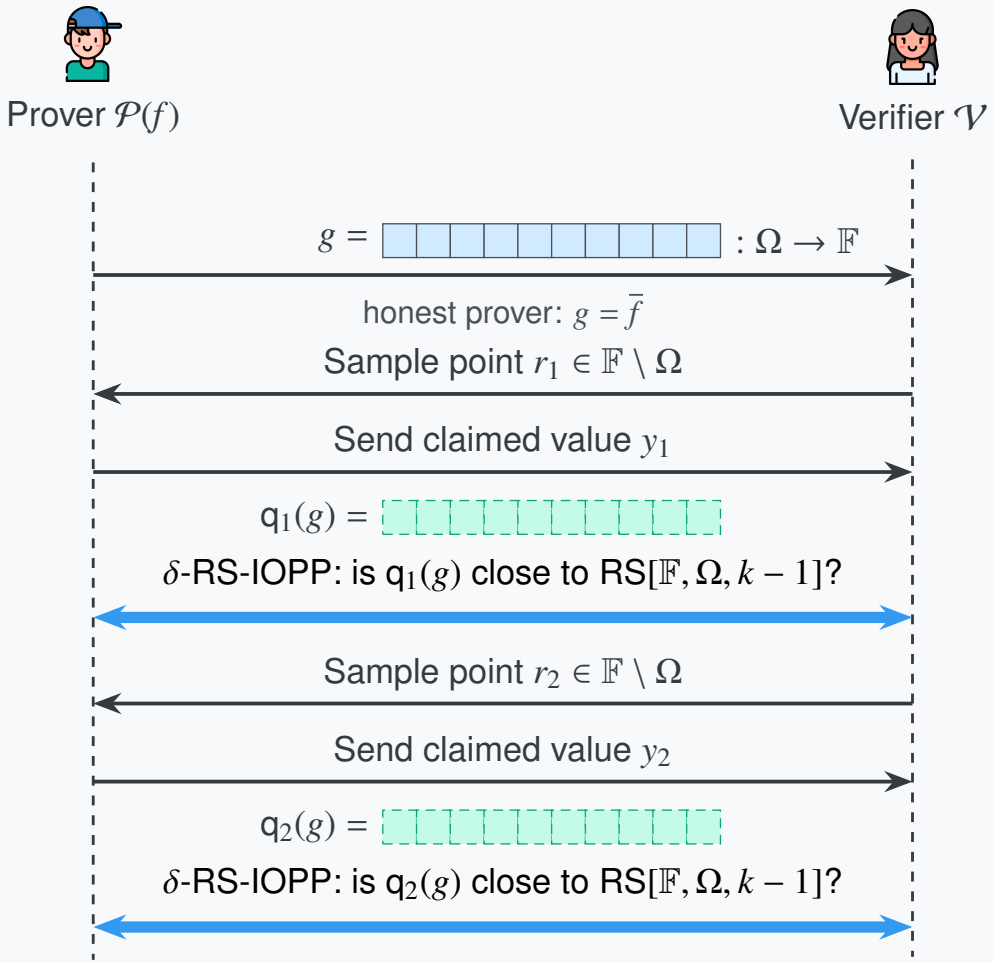


Figure: Virtual quotients $q_i(g) := (g - y_i)/(T - r_i)$: $\mathcal{V}$ computes their cells from g.

Attempt #1: What Goes Wrong?

What does the verifier $\mathcal{V}$ learn?

$\mathcal{V}$ learns

There are some $\bar{f}_1, \bar{f}_2 \in \mathsf{List}[g, k, \delta]$ s.t. $f_1(r_1) = y_1$ and $f_2(r_2) = y_2$.

Problem

For $\delta \geq \mu(C)/2$, the list $\mathsf{List}[g, k, \delta]$ may contain several elements, so it might happen that $f_1 \neq f_2$: different queries are explained by different codewords!

Attempt #2: Binding with a Random Point

Fix: before everything, $\mathcal{V}$ samples $r \leftarrow\$ \mathbb{F} \setminus \Omega$, and an honest prover responds with $y \leftarrow f(r)$.

Proposition

Let Bad be the event "there exist $\bar{f}_1 \neq \bar{f}_2 \in \mathsf{List}[g, k, \delta]$ such that $f_1(r) = f_2(r) = y$ for $r \leftarrow\$ \mathbb{F} \setminus \Omega$". Then

$$\Pr[\mathsf{Bad}] \leq \binom{\#\mathsf{List}[g, k, \delta]}{2} \cdot \frac{k-1}{\#\mathbb{F} - \#\Omega}.$$

Proof. For a fixed pair, $f_1 - f_2 \neq 0$ has degree $< k$, so by Schwartz–Zippel $\Pr[f_1(r) = f_2(r)] \leq \frac{k-1}{\#\mathbb{F}-\#\Omega}$. Union bound over all pairs. □

Attempt #2: Probability of the Bad Event

Corollary

If $\delta < 1 - \sqrt{\rho(C)}$ (for $C = \mathsf{RS}[\mathbb{F}, \Omega, k]$), then

$$\Pr[\text{Bad}] \leq \frac{1}{2\varepsilon_\delta^2} \cdot \frac{k-1}{\#\mathbb{F} - \#\Omega}, \quad \varepsilon_\delta = 2\sqrt{\rho(C)}(1 - \sqrt{\rho(C)} - \delta),$$

where ε_δ does not depend on the size of the field.

Proof. Johnson bound and $\binom{x}{2} \leq \frac{x^2}{2}$. □

Remark

The probability is roughly const/$\#\mathbb{F}$: negligible for large fields. For small fields (like `M31`, $\#\mathbb{F} = 2^{31} - 1$), repeat the procedure several times.

Hence (r, y) binds the prover to **only one** $\bar{f} \in \mathsf{List}[g, k, \delta]$. Remaining problem: aggregate the checks $f(r) = y$ and $f(r_i) = y_i$.

Multiple Quotientings

Proposition (Multiple Quotientings)

Let $(r_i)_{i\in[t]} \subseteq \mathbb{F} \setminus \Omega$, $(y_i)_{i\in[t]} \subseteq \mathbb{F}$, and $\delta \in [0,1]$. Define $\nu, \iota \in \mathbb{F}[T]^{\leq t}$:

$$\nu(T) := \prod_{i\in[t]}(T - r_i), \quad \iota(r_i) = y_i \text{ for all } i \in [t],$$

and the **quotient map** $\mathsf{q} : g \mapsto (g(T) - \iota(T))/\nu(T)$. Then:

1. if $g = \bar{f} \in \mathsf{RS}[\mathbb{F}, \Omega, k]$ and $y_i = f(r_i)$ for all $i \in [t]$, then $\mathsf{q}(g) \in \mathsf{RS}[\mathbb{F}, \Omega, k - t]$;
2. if for every $\bar{h} \in \mathsf{List}[g, k, \delta]$ we have $y_i \neq h(r_i)$ for some $i \in [t]$, then $\mathsf{q}(g)$ is δ-far from $\mathsf{RS}[\mathbb{F}, \Omega, k - t]$.

Question #1

Prove this fact (generalize the proof of single quotienting).

Attempt #2

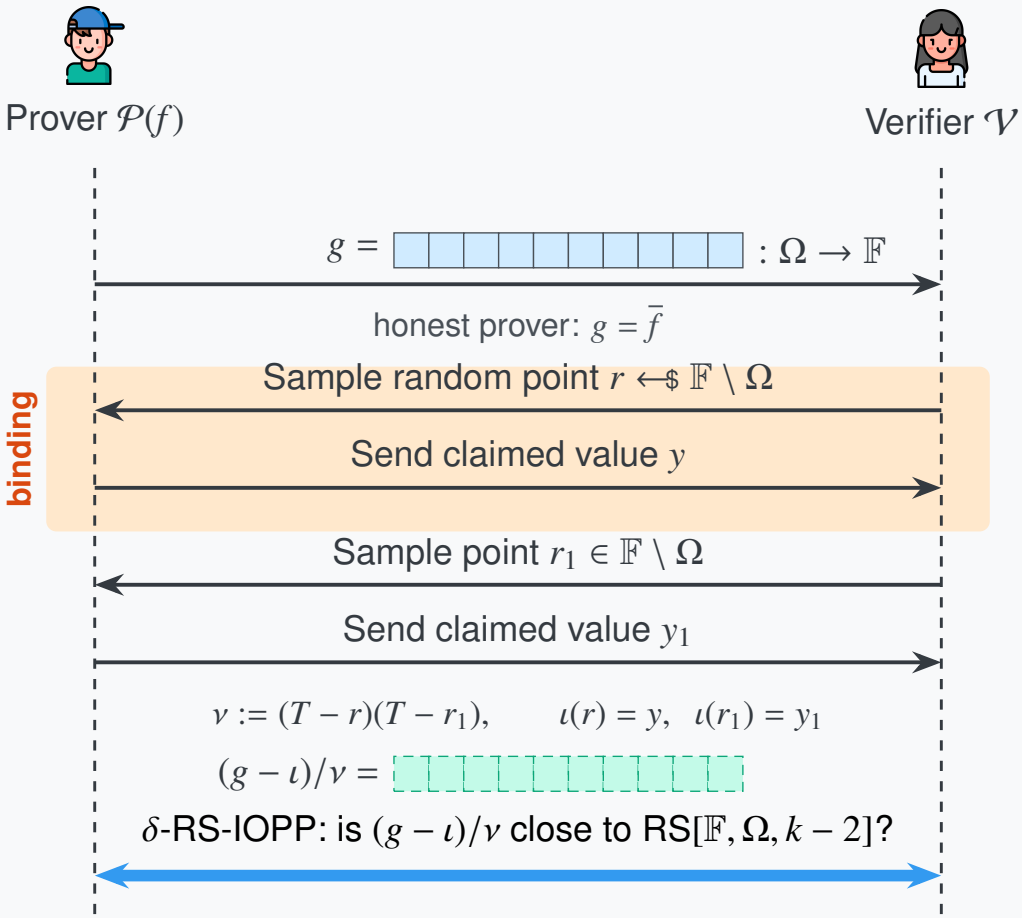


Figure: New: a random out-of-domain point r (highlighted); later queries are quotiented *together* with (r, y).

Batching RS-IOPPs

Batching RS-IOPPs

Running one RS-IOPP per query is costly. Given oracles $g_0, \ldots, g_n : \Omega \to \mathbb{F}$, the prover wants to convince $\mathcal{V}$ that all of them are δ-close to $\mathrm{RS}[\mathbb{F}, \Omega, k]$.

Definition (Distance Preserving Transformation)

Let $\Omega, \Omega' \subseteq \mathbb{F}$ be evaluation domains and r a blinding factor. The map $T : (g_0, \ldots, g_n; r) \mapsto g$, from restrictions $g_i : \Omega \to \mathbb{F}$ to $g : \Omega' \to \mathbb{F}$, is **distance preserving** if:

1. if $g_0, \ldots, g_n \in \mathrm{RS}[\mathbb{F}, \Omega, k]$, then $g \in \mathrm{RS}[\mathbb{F}, \Omega', k']$ for all r;
2. if some g_i is δ-far from $\mathrm{RS}[\mathbb{F}, \Omega, k]$, then g is δ-far from $\mathrm{RS}[\mathbb{F}, \Omega', k']$ with overwhelming probability over r.

Linear Combination

Example

For r sampled randomly from $\mathbb{F}$,

$$T(g_0, \ldots, g_n; r) := g_0 + g_1 r + g_2 r^2 + \cdots + g_n r^n$$

is a distance preserving transformation (for appropriate code parameters).

Batching: $\mathcal{V}$ samples $r \leftarrow\$ \mathbb{F}$ and the prover sets $g := \sum_i r^i g_i$. Then one RS-IOPP for g suffices.

- Property 1 is immediate: a linear combination of codewords is a codeword.
- Property 2 is tricky: it follows from the **proximity gap theorem**.

Proximity Gaps

Lemma (The proximity gap theorem, simplified)

Let $C := \mathsf{RS}[\mathbb{F}, \Omega, k]$, $g_0, \ldots, g_n : \Omega \to \mathbb{F}$, $\delta < 1 - 1.01\sqrt{\rho}$ and $g_r := \sum_i r^i g_i$. Suppose

$$\Pr_{r \leftarrow \$ \mathbb{F}} [\delta(g_r, C) \leq \delta] > \varepsilon_\delta,$$

where $\varepsilon_\delta = O(\frac{nk}{\rho \# \mathbb{F}})$ for $\delta < \frac{1-\rho}{2}$ and $\varepsilon_\delta = O(\frac{nk^2}{\rho^2 \# \mathbb{F}})$ otherwise. Then all g_i are δ-close to C.

Theorem (BCIKS20)

Moreover, there exists $S \subseteq \Omega$ such that $\#S \geq (1-\delta)\#\Omega$ and

$$\forall_i \; \exists_{f_i \in C} \quad g_i\big|_S = f_i\big|_S.$$

That is, all g_i are δ-close to C **on the same positions** S.

2-way Folding: Splitting a Polynomial

Another distance preserving transformation shrinks the **code** instead of the number of functions.

- For $f \in \mathbb{F}[T]^{<k}$, let f_E (f_O) consist of the even (odd) coefficients of f:

$$f(T) = f_E(T^2) + Tf_O(T^2), \quad f_E, f_O \in \mathbb{F}[T]^{<k/2}.$$

- Let $\Omega := \langle\omega\rangle$, where ω is a primitive nth root of unity, n a power of two.
- Since $\omega^{n/2} = -1$, Ω splits into pairs $\{x, -x\}$, and both elements of a pair are mapped to the same element of $\Omega^2 := \{x^2 : x \in \Omega\}$.

Question #2

Verify that $\#\Omega^2 = \frac{1}{2}\#\Omega$, so the evaluation domain shrinks $2\times$.

Squaring Halves a Smooth Domain

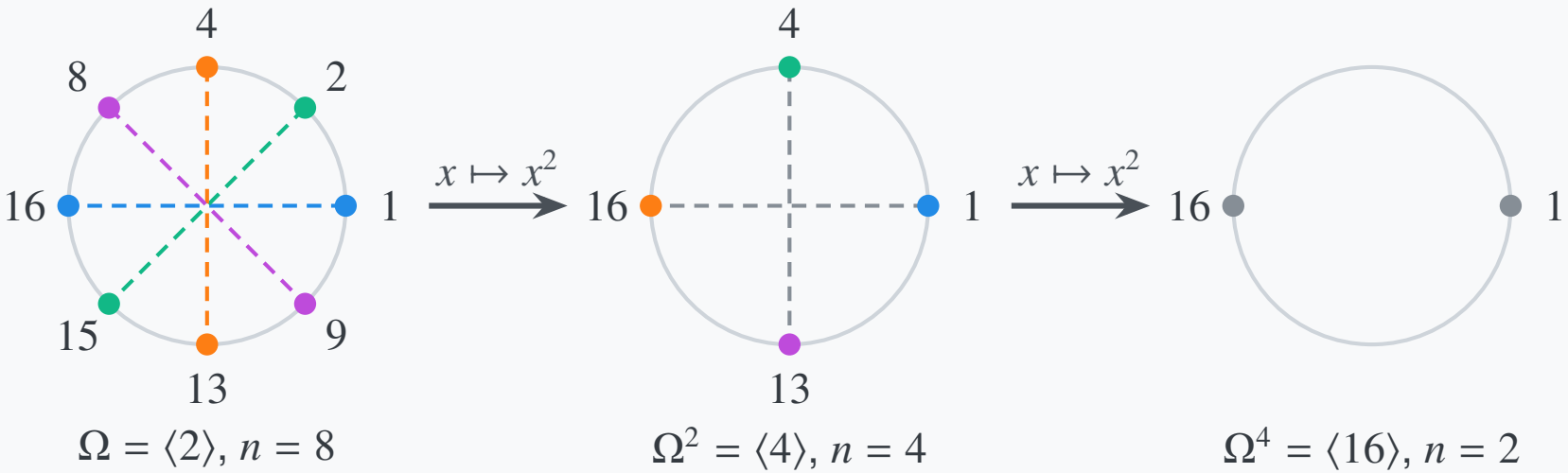


Figure: In $\mathbb{F}_{17}$: in $\Omega = \langle 2 \rangle$, every x has $-x$ on the opposite side (dashed), and both are mapped to the point of the same colour in Ω^2. Applying the map again gives $\Omega^4 = \{1, -1\}$.

2-way Folding

If $g = \bar{f}$, plugging x and $-x$ into $f(T) = f_E(T^2) + Tf_O(T^2)$ gives two linear equations:

$$f_E(x^2) = \frac{g(x) + g(-x)}{2}, \qquad f_O(x^2) = \frac{g(x) - g(-x)}{2x}.$$

The right-hand sides make sense for an **arbitrary** restriction g.

Definition (2-way folding)

For a restriction $g : \Omega \to \mathbb{F}$ and $r \in \mathbb{F}$, define $g_E, g_O : \Omega^2 \to \mathbb{F}$ by

$$g_E(x^2) := \frac{g(x) + g(-x)}{2}, \qquad g_O(x^2) := \frac{g(x) - g(-x)}{2x},$$

and the **fold** of g as $\mathrm{Fold}(g, r) := g_E + r \cdot g_O : \Omega^2 \to \mathbb{F}$.

Folding is Distance Preserving

Lemma

Let $C := \mathrm{RS}[\mathbb{F}, \Omega, k]$, $C' := \mathrm{RS}[\mathbb{F}, \Omega^2, k/2]$, and δ, ε_δ as in the proximity gap lemma. Then:

1. if $g = \overline{f} \in C$, then $\mathrm{Fold}(g, r) = \overline{f_E + r f_O} \in C'$ for all $r \in \mathbb{F}$;
2. if g is δ-far from C, then $\Pr_{r \leftarrow \$ \mathbb{F}}[\mathrm{Fold}(g, r) \text{ is } \delta\text{-close to } C'] \leq \varepsilon_\delta$.

Proof idea of (ii). $\mathrm{Fold}(g, r)$ is a random linear combination of g_E, g_O; C, C' have the same rate.

- The theorem gives $S \subseteq \Omega^2$ and $u_E, u_O \in \mathbb{F}[T]^{<k/2}$ agreeing with g_E, g_O on S.
- Glue: $u(T) := u_E(T^2) + T \cdot u_O(T^2)$ agrees with g on the $2 \cdot \#S \geq (1 - \delta)n$ preimages of S: contradiction. □

Remark

We need the **strong** form: the **same** set S for g_E and g_O.

Attempt #3: Putting Everything Together

Attempt #3: The Full Compilation

The Poly-IOP prover sends $f_1, \ldots, f_m \in \mathbb{F}[T]^{<k}$; the Poly-IOP verifier wants all of them at $r_1, \ldots, r_t \in \mathbb{F} \setminus \Omega$.

1: **Commit.** $\mathcal{P}$ sends oracles $g_1, \ldots, g_m : \Omega \to \mathbb{F}$; honestly, $g_j := \overline{f_j}$.
2: **Bind.** $\mathcal{V}$ samples $r_0 \leftarrow\!\$\ \mathbb{F} \setminus \Omega$, $\mathcal{P}$ replies with $y_{j,0}$ $(= f_j(r_0))$ for every $j \in [m]$.
3: **Run the Poly-IOP.** When asked for $f_j(r_i)$, $\mathcal{P}$ replies with $y_{j,i}$, and $\mathcal{V}$ simply uses it.
4: **Quotient.** $\nu(T) := \prod_{i=0}^{t}(T - r_i)$, $\iota_j \in \mathbb{F}[T]^{\leq t}$ with $\iota_j(r_i) = y_{j,i}$, and

$$\mathsf{q}_j := \frac{g_j - \iota_j}{\nu} : \Omega \to \mathbb{F}, \quad j \in [m].$$

5: **Batch.** $\mathcal{V}$ samples $\alpha \leftarrow\!\$\ \mathbb{F}$; both define $h := \sum_{j\in[m]} \alpha^j \mathsf{q}_j$.
6: **Test.** Run a single δ-RS-IOPP for h against $\mathsf{RS}[\mathbb{F}, \Omega, k - t - 1]$.

Attempt #3

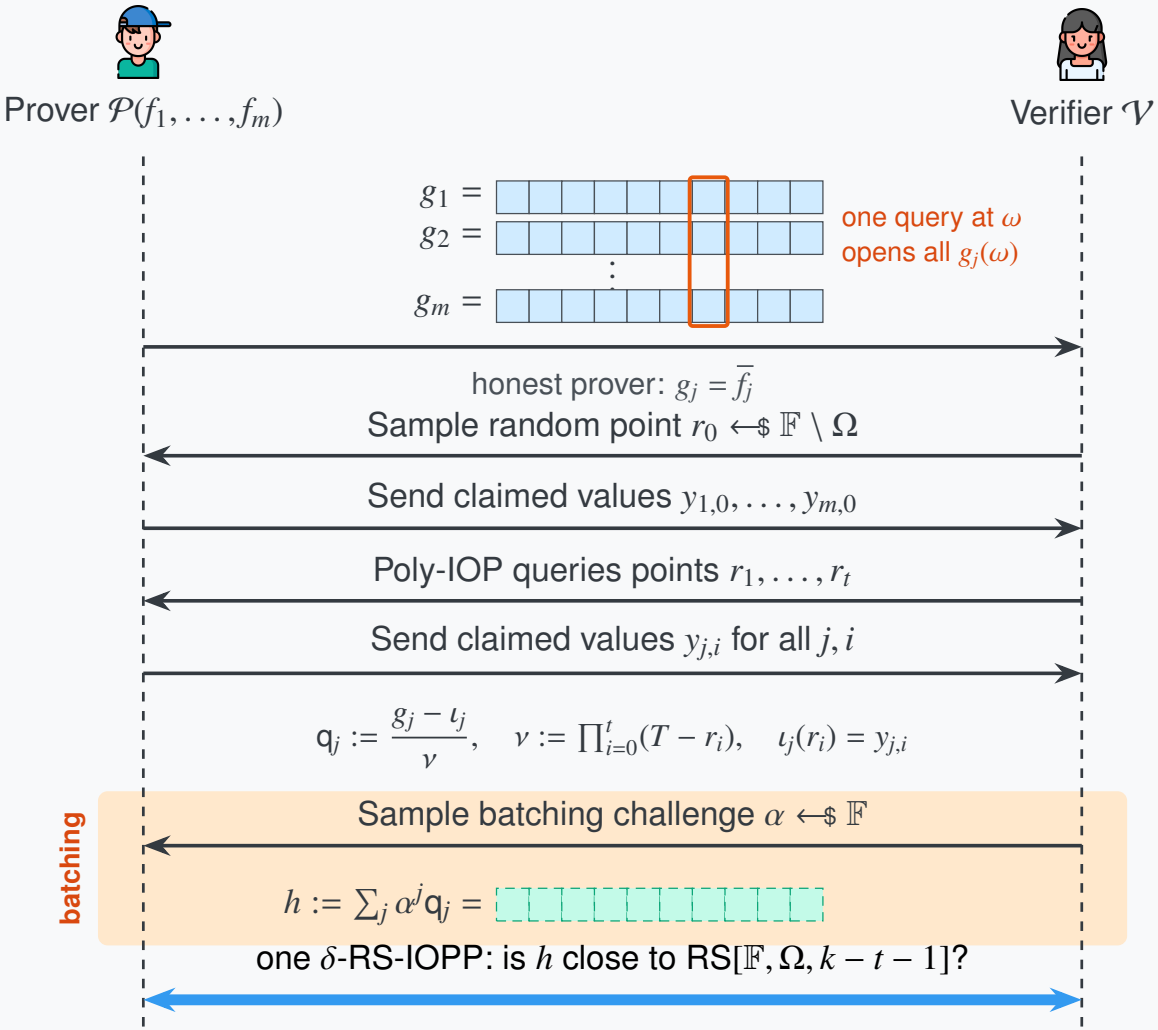


Figure: Commit, bind, answer, quotient, batch with α (highlighted), and run only one RS-IOPP.

Attempt #3: Why it Works

Virtual oracles. No oracles for q_j or h: $\mathcal{V}$ queries $g_1(\omega), \ldots, g_m(\omega)$ and computes $h(\omega)$.

Completeness. Every $\mathsf{q}_j \in \mathsf{RS}[\mathbb{F}, \Omega, k - t - 1]$, hence so is h.

Soundness. Suppose $\mathcal{V}$ accepts with non-negligible probability. Then:

1: h is δ-close to the code, so by proximity gaps every q_j is δ-close to $\mathsf{RS}[\mathbb{F}, \Omega, k - t - 1]$ (except w.p. ε_δ over α);

2: by multiple quotienting, each g_j is δ-close to some $\overline{f'_j}$ with $f'_j(r_i) = y_{j,i}$ for **all** i;

3: $\overline{f'_j}$ is the only codeword in $\mathsf{List}[g_j, k, \delta]$ through $(r_0, y_{j,0})$, except w.p. $\Pr[\mathsf{Bad}]$ (union bound over j).

All answers come from one fixed f'_j per oracle: exactly what a polynomial commitment must guarantee.

Degree Correction

Remark

If different polynomials are queried at different numbers of points, the quotients q_j have different degree bounds and cannot be added directly.

Degree correction

Before batching, each q_j is multiplied by a power of T (and combined with fresh random coefficients), so that all of them are tested against the same code.

Question #3

Why do we sample r_0 from $\mathbb{F} \setminus \Omega$ and not from Ω? What would go wrong with the quotient q_j otherwise?

Summary

- An **IOPP** gives the verifier only oracle access to the input $\mathbb{y}$; we can only ask it to reject inputs that are δ**-far** from valid ones.
- A δ**-RS-IOPP** tests proximity of $g : \Omega \to \mathbb{F}$ to RS$[\mathbb{F}, \Omega, k]$.
- **Quotienting** turns the claim $f(r) = y$ into proximity of $(g - y)/(T - r)$ to a smaller RS code.
- **Out-of-domain sampling** at $r_0 \leftarrow\$ \mathbb{F} \setminus \Omega$ binds the prover to a single codeword of the list.
- **Proximity gaps**: random linear combinations and **folding** are distance preserving, so all claims are batched into one RS-IOPP.
- Next: building the RS-IOPP itself (FRI) and the STARK.

Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



Blockstream